

Vacuum Cleaner Problem

Aim

To implement the **Vacuum Cleaner Problem** in Python, where the vacuum cleaner cleans all dirty rooms by moving between them.

Algorithm

1. Initialize the environment with two rooms (A and B).
2. Check the current room.
3. If the room is dirty, clean it.
4. Move to the other room.
5. Repeat the process until both rooms are clean.
6. Display the final state of the rooms.

Python Code

```
# Vacuum Cleaner Problem
```

```
rooms = {  
    "A": "Dirty",  
    "B": "Dirty"  
}  
  
current_room = "A"  
  
while True:  
    print(f'Vacuum is in Room {current_room}')  
  
    if rooms[current_room] == "Dirty":  
        print(f'Cleaning Room {current_room}')  
        rooms[current_room] = "Clean"  
    else:  
        print(f'Room {current_room} is already Clean')
```

```

# Check if all rooms are clean

if all(state == "Clean" for state in rooms.values()):
    break

# Move to the other room
current_room = "B" if current_room == "A" else "A"

print("\nFinal State:")

for room, state in rooms.items():
    print(f"Room {room}: {state}")

```

Output

```

.PY
Vacuum is in Room A
Cleaning Room A
Vacuum is in Room B
Cleaning Room B

Final State:
Room A: Clean
Room B: Clean
>>> |

```

Result

The **Vacuum Cleaner Problem** was successfully implemented using Python. The vacuum cleaner cleaned all the dirty rooms and reached the goal state where every room was clean.